

PROFESSIONAL EXPERIENCE

Research Lead <i>Tutor Intelligence</i> Foundation model research and training	Jun 2026 — Present Watertown, MA
Research Lead Research Scientist <i>RAI Institute (formerly Boston Dynamics AI Institute)</i> - Founded two research teams from scratch: Foundation Models (10+ researchers) and Capture (30 people); joined as employee #20 and helped grow organization to 275+ - Built 320 H100 GPU cluster for large-scale training of video prediction and multimodal foundation models - Lead the largest data collection effort at the institute with a roadmap for 100,000+ demonstrations; created task definition frameworks and benchmark protocols for data quality and consistency across teams - Built video prediction system for robot policies using internet-scale video as prior (early 2023) - Designed multimodal architectures combining diverse sensor modalities with internet-scale pretrained priors - Developed gradient-free RL enabling online learning with non-differentiable semantic reward functions (U.S. Patent pending) - Established research partnerships with Google, Columbia, ETH Zurich, Agile Robots	Sep 2023 — Jun 2026 Dec 2022 — Aug 2023 Cambridge, MA
Research Fellow <i>Army Research Lab</i> - Won 1st place MineRL BASALT Competition (NeurIPS 2021): combined imitation learning, human preference data, and hierarchical knowledge engineering to solve tasks defined only by natural language - Developed mobile manipulation approaches without runtime localization (IROS 2022 Best Paper Finalist) - Developed multimodal geometric learning fusing vision and tactile sensing for robotic grasping (ICRA 2019)	Sep 2018 — Jul 2022 Aberdeen, MD
CEO / Co-Founder <i>Odefi Inc. — Columbia IBM Blockchain Accelerator</i> Founded startup to build decentralized financial products on MakerDAO; drove technical architecture and investor pitches	Mar 2019 — Jan 2022 New York, NY

SELECTED RESEARCH

Theia: Vision Foundation Models for Robotics <i>CoRL 2024</i> - Developed spatial token distillation framework combining CLIP, DINOv2, SigLIP, and SAM into a single compact model - Achieved state-of-the-art on CortexBench and real-world robotics tasks with reduced model size; released open-source models	2024 Cambridge, MA
Preference-Based RL from Human Feedback <i>ICML 2023, NeurIPS 2021</i> - Developed DIP-RL, an algorithm that leverages human demonstrations to infer preferences and learn reward functions for RL agents in open-ended environments (ICML 2023) - Won 1st place and most human-like agent at MineRL BASALT (NeurIPS 2021) for learning from human feedback	2021 — 2023 New York, NY

SELECTED PUBLICATIONS (10 PEER-REVIEWED)

- Tellex & **Watkins**. “Elephants Don’t Write Sonnets: The Grounded Turing Test for Embodied AI.” *MIT Press* 2026
- Shang, Schmeckpeper, May, Minniti, Kelestemur, **Watkins**, Herlant. “Theia: Distilling Diverse Vision Foundation Models for Robot Learning.” *CoRL 2024*
- Cohen, Liu, Mooney, Tellex, **Watkins**. “A Survey of Robotic Language Grounding.” *IJCAI 2024*
- Novoseller, Goecks, **Watkins**, Miller, Waytowich. “DIP-RL: Demonstration-Inferred Preference Learning.” *ICML 2023*
- Watkins-Valls**, Maia, Varley, Seshadri, Sanabria, Waytowich, Allen. “Mobile Manipulation Leveraging Multiple Views.” *IROS 2022* (Best Paper Finalist)

EDUCATION

Ph.D. in Computer Science , <i>Columbia University</i> Thesis: <i>Learning Mobile Manipulation</i> Advisor: Prof. Peter Allen Army Research Lab Fellow	2017 — 2022
B.S./M.S. in Computer Science , <i>Columbia University</i> , 4.0 GPA (M.S.)	2012 — 2017

SKILLS

ML / AI Systems	Foundation Models, Robot Manipulation, Imitation Learning, Diffusion Policies, RLHF
Communication	Python, C++, PyTorch, CUDA, ROS 2, Distributed Training (320 H100s), GPU Infrastructure
	English, Spanish